SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Product form : Substance (UVCB)
Trade name : Sisterna SP70-C
REACH registration No. : 01-2120769127-48-XXXX
Type of product : Non ionic surfactant

1.2. Relevant identified uses of the substance or mixture and uses advised against

1.2.1. Relevant identified uses

Intended for general public
Main use category : Industrial use,Professional use,Consumer use
Industrial/Professional use spec : Industrial
Use of the substance/mixture : Non ionic surfactant
Function or use category : Surfactants

Title	Life cycle stage	Use descriptors
Hairdressing and other beauty treatment	Professional	PC39, PROC5, PROC8a, ERC8a
Cosmetics	Consumer	PC39, ERC8a
Ink	Consumer	PC18, ERC9a, ERC9b
Manufacture of cosmetics	Manufacture	PC39, PROC1, PROC2, PROC3, PROC5, PROC8a, PROC8b, PROC9, PROC14, PROC15, ERC2
Manufacture of paints, varnishes and similar coatings, printing ink and mastics	Manufacture	PC18, PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC9, ERC2

Full text of use descriptors: see section 16

1.2.2. Uses advised against

No additional information available

1.3. Details of the supplier of the safety data sheet

Supplier
Company : SISTERNA B.V.
Address : Belder 30A, 4704 RK Roosendaal, The Netherlands
Telephone number : +31-165-524730
E-mail address : info@sisterna.com

Name of manufacturer in Japan:

Manufacturer : DKS Co. Ltd.
Address : 5 Ogawara-cho, Kisshoin, Minami-ku, Kyoto 601-8391, Japan
Telephone number : +81-3-3275-0569 (Tokyo, Japan)
+81-75-326-7553 (Kyoto, Japan)
Fax number : +81-3-3275-0599 (Tokyo, Japan)
+81-75-326-7552 (Kyoto, Japan)
E-mail address : d-kankyou@dks-web.co.jp

Only representative: SCAS Europe S.A./N.V.

1.4. Emergency telephone number

Country	Organisation/Company	Address	Emergency number	Comment
England and Wales	NHS Direct in England or Wales		0845 46 47 or 111	

Country	Organisation/Company	Address	Emergency number	Comment
Ireland	Republic of Ireland		01 809 2166 (public 8am-10pm 7/w) 01 809 2566 (healthcare professionals 24/7)	
Scotland	NHS 24 in Scotland		08454 24 24 24 or 111	
South Africa	Red Cross Children's Hospital Poisons Information Centre	Room 411 ICH Building Red Cross Children's Hospital Klipfontein Rd 7700 Rondebosch	+27 (0)861 555 777	
USA	The American Association of Poison Control Centers (AAPCC)		1-800-222-1222 - 24/24	

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Serious eye damage/eye irritation, Category 2 H319
Full text of H- and EUH-statements: see section 16

Adverse physicochemical, human health and environmental effects

No additional information available

2.2. Label elements

Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms (CLP) :



GHS07

Signal word (CLP) : Warning
Hazard statements (CLP) : H319 - Causes serious eye irritation.
Precautionary statements (CLP) : P264 - Wash hands thoroughly after handling.
P280 - Wear eye protection, face protection.
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337+P313 - If eye irritation persists: Get medical advice/attention.

2.3. Other hazards

This substance/mixture does not meet the PBT criteria of REACH regulation, annex XIII
This substance/mixture does not meet the vPvB criteria of REACH regulation, annex XIII
Contains no PBT/vPvB substances ≥ 0.1% assessed in accordance with REACH Annex XIII

Component	
Fatty acids C16-18 (even numbered), mono and di esters with sucrose	This substance/mixture does not meet the PBT criteria of REACH regulation, annex XIII This substance/mixture does not meet the vPvB criteria of REACH regulation, annex XIII

The substance is not included in the list established in accordance with Article 59(1) of REACH for having endocrine disrupting properties, or is not identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605

SECTION 3: Composition/information on ingredients

3.1. Substances

Substance type : UVCB
Name : Fatty acids C16-18 (even numbered), mono and di esters with sucrose

Name	Product identifier	%
Fatty acids C16-18 (even numbered), mono and di esters with sucrose	REACH-no: 01-2120769127-48-XXXX	100

Full text of H- and EUH-statements: see section 16

3.2. Mixtures

Not applicable

SECTION 4: First aid measures

4.1. Description of first aid measures

First-aid measures general : No action shall be taken without appropriate training or involving any personal risk. Mouth-to-mouth resuscitation is forbidden (possible poisoning of first-aider).

First-aid measures after inhalation : Remove person to fresh air and keep comfortable for breathing. In case of irregular breathing or respiratory arrest provide artificial respiration. If experiencing respiratory symptoms: Call a poison center or a doctor. Do not apply mouth-to-mouth resuscitation. If unconscious place in recovery position and seek medical advice.

First-aid measures after skin contact : After contact with skin, wash immediately and thoroughly with polyethylene glycol, followed by plenty of water. Wash immediately with plenty of soap and water. Remove contaminated clothing and shoes. Get medical attention if irritation persists after washing. Wash contaminated clothing before reuse. Thoroughly clean shoes before re-using.

First-aid measures after eye contact : In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Remove contact lenses if easy to do. Immediately rinse with water for a prolonged period while holding the eyelids wide open. Get immediate medical advice/attention. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.

First-aid measures after ingestion : Rinse mouth out with water. Remove person to fresh air and keep comfortable for breathing. If the person is fully conscious, make him/her drink water. Never give an unconscious person anything to drink. Do not induce vomiting without medical advice. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Never give anything by mouth to an unconscious person. If unconscious place in recovery position and seek medical advice. Loosen tight clothing such as a collar, tie, belt or waistband.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms/effects : Redness of the eye tissue. Eye irritation. Irritation of the respiratory tract. Cough.
Symptoms/effects after inhalation : Dust of the product, if present, may cause respiratory irritation after an excessive inhalation exposure.
Symptoms/effects after skin contact : No known significant effects or critical hazards.
Symptoms/effects after eye contact : Causes serious eye irritation.
Symptoms/effects after ingestion : No known significant effects or critical hazards.

4.3. Indication of any immediate medical attention and special treatment needed

Treat symptomatically. On ingestion in large quantities: Call a physician immediately.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media : Dry chemical, CO2, or water spray or regular foam. Use extinguishing agent suitable for surrounding fire.
Unsuitable extinguishing media : Do not use water jet.

5.2. Special hazards arising from the substance or mixture

Fire hazard : May form explosive dust-air mixture if dispersed.
Hazardous decomposition products in case of fire : Carbon dioxide. Carbon monoxide.

5.3. Advice for firefighters

Firefighting instructions : Evacuate danger area. No action shall be taken without appropriate training or involving any personal risk. Move containers from fire area if it can be done without personal risk. Use water spray or fog for cooling exposed containers.
Protection during firefighting : Wear recommended personal protective equipment. Use breathing equipment (SCBA). Do not enter fire area without proper protective equipment, including respiratory protection. Wear fire/flame resistant/retardant clothing. Standard EN 469 - Protective clothing for firefighters. Standard - EN 659: Protective gloves for firefighters. Use self-contained breathing apparatus and chemically protective clothing.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

6.1.1. For non-emergency personnel

Protective equipment : Wear recommended personal protective equipment.
Emergency procedures : No action shall be taken without appropriate training or involving any personal risk. Evacuate the danger area. Evacuate unnecessary personnel. Do not touch or walk on the spilled product. Remove all sources of ignition. No open flames. No smoking. No flames. Eliminate all sources of ignition. Avoid breathing (dust, vapor, mist, gas). Ensure adequate ventilation. In case of inadequate ventilation wear respiratory protection. Use appropriate personal protection equipment (PPE).

6.1.2. For emergency responders

Protective equipment : For further information refer to section 8: "Exposure controls/personal protection". Wear recommended personal protective equipment.
Emergency procedures : See Section 8 for information on personal protection equipment.

6.2. Environmental precautions

Take all necessary measures to avoid accidental discharge of products into drains and waterways due to the rupture of containers or transfer systems.

6.3. Methods and material for containment and cleaning up

For containment : Small spills: Move containers from spill area. Use only spark proof tools. Use explosion-proof equipment. Avoid dust formation. Use of a vacuum with high efficiency particulate air (HEPA) filtration is recommended. Place in an appropriate container and dispose of the contaminated material at a licensed site. In case of large spillages: Move containers from spill area. Use explosion-proof equipment. Use only spark proof tools. Use of a vacuum with high efficiency particulate air (HEPA) filtration is recommended. Prevent entry to sewers and public waters. Approach from upwind. Minimize airborne dust generation and prevent wind dispersal during loading and unloading. Place in an appropriate container and dispose of the contaminated material at a licensed site.

6.4. Reference to other sections

For further information see section 1. For further information refer to section 8: "Exposure controls/personal protection". For further information refer to section 13.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling : Use appropriate personal protection equipment (PPE). Do not ingest. Avoid contact with skin, eyes and clothing. Avoid ignition sources. Avoid dust formation. Prevent dust accumulation (to minimize explosion hazard). Provide local exhaust or general room ventilation. Wear appropriate breathing apparatus if air renewal not sufficient to maintain dust/vapour under TLV. Electrical equipment should be protected to the appropriate standard. Take precautionary measures against static discharge. The product may charge electrostatically: use earthling leads when transferring from one container to another. Empty containers retain product residue and can be hazardous.

Hygiene measures : Do not eat, drink or smoke when using this product. Wash hands before work breaks and after finishing work. Remove contaminated clothing and protective equipment before entering eating areas. For further information refer to section 8: "Exposure controls/personal protection".

7.2. Conditions for safe storage, including any incompatibilities

Storage conditions : Store in accordance with local regulations. Store in original container. Store away from direct sunlight or other heat sources. Store in a well-ventilated place. Keep cool. Refer to Section 10 on Incompatible Materials. Protect from sunlight. Keep away from ignition sources. Keep container tightly closed. Do not store near oxidizing agents. Containers which are opened must be carefully resealed and kept upright to prevent leakage. do not store in unlabelled containers. Use appropriate container to avoid environmental contamination. For further information, refer to section 10 : "Stability and Reactivity".

7.3. Specific end use(s)

See Section 7 for information on safe handling. For further information see section 1.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

8.1.1 National occupational exposure and biological limit values

No additional information available

8.1.2. Recommended monitoring procedures

Monitoring methods	
Monitoring methods	Workplace atmospheres. Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy. Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. Workplace exposure - General requirements for the performance of procedures for the measurement of chemical agents.

8.1.3. Air contaminants formed

No additional information available

8.1.4. DNEL and PNEC

Sisterna SP70-C	
PNEC (Water)	
PNEC aqua (freshwater)	0,003 mg/l Assessment Factors
PNEC aqua (marine water)	0,003 mg/l Assessment Factors
PNEC (Sediment)	
PNEC sediment (freshwater)	3,979 mg/kg dwt Equilibrium Partitioning
PNEC sediment (marine water)	0,398 mg/kg dwt Equilibrium Partitioning

Sisterna SP70-C	
PNEC (Soil)	
PNEC soil	0,794 mg/kg dwt Assessment Factors
PNEC (STP)	
PNEC sewage treatment plant	10 mg/l Assessment Factors

8.1.5. Control banding

No additional information available

8.2. Exposure controls

8.2.1. Appropriate engineering controls

Appropriate engineering controls:
Ensure that there is a suitable ventilation system. If handling results in dust generation or high temperatures, local exhaust ventilation should be provided to insure that exposure to dust or decomposition products does not exceed the exposure recommended levels. Use explosion-proof electrical, ventilating and lighting equipment.

8.2.2. Personal protection equipment

Personal protective equipment symbol(s):



8.2.2.1. Eye and face protection

Eye protection:
Wear closed safety glasses

Eye protection			
Type	**Field of application**	**Characteristics**	**Standard**
Safety goggles, Safety glasses	Droplet, Dust, Fine dust		EN 166

8.2.2.2. Skin protection

Skin and body protection:
Protective clothing

Hand protection:
Protective gloves against chemicals (EN 374)

Hand protection					
Type	**Material**	**Permeation**	**Thickness (mm)**	**Penetration**	**Standard**
Reusable gloves					EN ISO 374

Other skin protection

Materials for protective clothing:
Wear foot protection. Safety footwear

8.2.2.3. Respiratory protection

Respiratory protection:
Wear appropriate mask

Respiratory protection			
Device	**Filter type**	**Condition**	**Standard**
Disposable half mask	Type P2		EN 14387

8.2.2.4. Thermal hazards

No additional information available

8.2.3. Environmental exposure controls

Environmental exposure controls:
Ensure that the emission levels from local regulations or operating permits are not exceeded. Use engineering controls to keep exposures below the OEL or DNEL.

Other information:
Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work. Take off contaminated clothing and wash it before reuse. Emergency safety showers should be available in the immediate vicinity of any potential exposure.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	:	Solid
Colour	:	white. light yellow. light brown.
Appearance	:	Powder.
Odour	:	characteristic. slight.
Odour threshold	:	Not available
Melting point	:	58 °C Approximately
Freezing point	:	Not applicable
Boiling point	:	Not applicable
Flammability	:	Non flammable.
Lower explosion limit	:	60 g/m³
Upper explosion limit	:	Not applicable
Flash point	:	215 °C (closed cup)
Auto-ignition temperature	:	Not applicable
Decomposition temperature	:	Not available
pH	:	7,8 Approximately
pH solution	:	Not available
Viscosity, kinematic	:	Not applicable
Solubility	:	Partially soluble. Soluble at 60°C. Soluble in water.
Partition coefficient n-octanol/water (Log Kow)	:	Not available
Partition coefficient n-octanol/water (Log Pow)	:	> 5,8 Calculation method
Vapour pressure	:	Not available
Vapour pressure at 50°C	:	Not available
Density	:	Not available
Relative density	:	0,45 Approximately
Relative vapour density at 20°C	:	Not applicable
Particle size	:	Not available

9.2. Other information

9.2.1. Information with regard to physical hazard classes

No additional information available

9.2.2. Other safety characteristics

No additional information available

SECTION 10: Stability and reactivity

10.1. Reactivity

No specific data.

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

None under normal use.

10.4. Conditions to avoid

Avoid dust formation. Avoid ignition sources. Avoid open fire or flames. Prevent the build-up of electrostatic charge. The product may charge electrostatically: use earthling leads when transferring from one container to another. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

10.5. Incompatible materials

Oxidizing agent.

10.6. Hazardous decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

11.1.1. Acute toxicity

Acute toxicity (oral) : Not classified (Based on available data, the classification criteria are not met)
Acute toxicity (dermal) : Not classified
Acute toxicity (inhalation) : Not classified

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
LD50 oral rat	> 2000 mg/kg (OECD 401 method)

11.1.2. Skin corrosion/irritation

Skin corrosion/irritation : Not classified (Based on available data, the classification criteria are not met)
pH: 7,8 Approximately

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Skin corrosion/irritation, In vivo, rabbit	Not irritating (OECD 404 method)

11.1.3. Serious eye damage/irritation

Serious eye damage/irritation : Causes serious eye irritation.
pH: 7,8 Approximately

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Serious eye damage/irritation, In vivo, rabbit	Irritating (OECD 405 method)
Serious eye damage/irritation, In vitro	Not irritating (OECD 437 method)
Serious eye damage/irritation, In vitro, chicken	Not irritating (OECD 438 method)

11.1.4. Respiratory or skin sensitisation

Respiratory or skin sensitisation : Not classified (Based on available data, the classification criteria are not met)

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Skin sensitization, skin, Guinea pig	Did not cause sensitisation (OECD 406 method)

11.1.5. Germ cell mutagenicity

Germ cell mutagenicity : Not classified (Based on available data, the classification criteria are not met)

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Germ cell mutagenicityIn vitro, Bacteria	Negative (OECD 471 method)

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Germ cell mutagenicityIn vitro, mammalian	Negative (OECD 490 method)
Germ cell mutagenicityIn vivo, mammalian	Negative (OECD 474 method)

11.1.6. Carcinogenicity

Carcinogenicity : Not classified
Additional information : No known significant effects or critical hazards

11.1.7. Reproductive toxicity

Reproductive toxicity : Not classified
Additional information : Based on available data, the classification criteria are not met

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Reproductive toxicity, Fertility, Developmental toxicity, (OECD 421 method)	NOAEL (oral, rat) 6000 mg/kg bw/day

11.1.8. STOT-single exposure

STOT-single exposure : Not classified

11.1.9. STOT-repeated exposure

STOT-repeated exposure : No known significant effects or critical hazards (Based on available data, the classification criteria are not met)

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
NOAEL (subchronic, oral, animal/male, 90 days)	3240 mg/kg bodyweight (OECD 408 method)
NOAEL (subchronic, oral, animal/female, 90 days)	3430 mg/kg bodyweight (OECD 408 method)

11.1.10. Aspiration hazard

Aspiration hazard : Not classified

11.2. Information on other hazards

No additional information available

SECTION 12: Ecological information

12.1. Toxicity

Hazardous to the aquatic environment, short–term (acute) : Not classified (Based on available data, the classification criteria are not met)
Hazardous to the aquatic environment, long–term (chronic) : Not classified
Not rapidly degradable

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
LC50 - Fish [1]	> 3,3 mg/l (OECD 203 method) - Danio Rerio
EC50 - Crustacea [1]	> 100 mg/l (OECD 202 method) Daphnia Magna
EC50 - Other aquatic organisms [1]	> 1000 mg/l (OECD 209 method)
EC50 72h - Algae [1]	> 100 mg/l (OECD 201 method) - Pseudokirchneriella subcapitata
EC50 72h - Algae [2]	> 100 mg/l (OECD 201 method) - Pseudokirchneriella subcapitata

12.2. Persistence and degradability

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Persistence and degradability	Readily biodegradable.
Biodegradation, (OECD 301C method)	> 85% (28 days)

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Biodegradation, (OECD 301B method)	> 54% (28 days)

12.3. Bioaccumulative potential

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Partition coefficient n-octanol/water (Log Pow)	> 5,8 Calculation method

12.4. Mobility in soil

Fatty acids C16-18 (even numbered), mono and di esters with sucrose	
Organic Carbon Normalized Adsorption Coefficient (Log Koc)	12022,64 at 20 °C
Additional information	Quantitative structure-activity relationship (QSAR)

12.5. Results of PBT and vPvB assessment

Fatty acids C16-18 (even numbered), mono and di esters with sucrose
This substance/mixture does not meet the PBT criteria of REACH regulation, annex XIII
This substance/mixture does not meet the vPvB criteria of REACH regulation, annex XIII

Component	
Fatty acids C16-18 (even numbered), mono and di esters with sucrose	This substance/mixture does not meet the PBT criteria of REACH regulation, annex XIII This substance/mixture does not meet the vPvB criteria of REACH regulation, annex XIII

12.6. Endocrine disrupting properties

Adverse effects on the environment caused by endocrine disrupting properties : No known significant effects or critical hazards.

12.7. Other adverse effects

Additional information : No known significant effects or critical hazards

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Waste treatment methods : The generation of waste should be avoided or minimized wherever possible. External treatment and disposal of waste should comply with applicable local and/or national regulations. Dispose of contents/container in accordance with licensed collector's sorting instructions. Dispose of at a licensed waste collection centre. Must follow special treatment according to local regulation. The allocation of waste identity numbers/waste descriptions must be carried out according to the EEC, specific to the industry and process.

Product/Packaging disposal recommendations : The generation of waste should be avoided or minimized wherever possible.

Additional information : This material and its container must be disposed of in a safe way, and as per local legislation. Care should be taken when handling emptied containers that have not been thoroughly cleaned or rinsed out. Empty containers retain product residue and can be hazardous. Take all necessary measures to avoid accidental discharge of products into drains and waterways due to the rupture of containers or transfer systems.

SECTION 14: Transport information

In accordance with ADR / IMDG / IATA / ADN / RID

ADR	IMDG	IATA	ADN	RID
14.1. UN number or ID number				
Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
14.2. UN proper shipping name				
Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
14.3. Transport hazard class(es)				
Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
14.4. Packing group				
Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
14.5. Environmental hazards				
Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
No supplementary information available				

14.6. Special precautions for user

Overland transport
Not applicable

Transport by sea
Not applicable

Air transport
Not applicable

Inland waterway transport
Not applicable

Rail transport
Not applicable

14.7. Maritime transport in bulk according to IMO instruments

Not applicable

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. EU-Regulations

REACH Annex XVII (Restriction List)

Not listed on REACH Annex XVII

REACH Annex XIV (Authorisation List)

Not listed on REACH Annex XIV (Authorisation List)

REACH Candidate List (SVHC)

Not listed on the REACH Candidate List

PIC Regulation (Prior Informed Consent)

Not listed on the PIC list (Regulation EU 649/2012)

POP Regulation (Persistent Organic Pollutants)

Not listed on the POP list (Regulation EU 2019/1021)

Ozone Regulation (1005/2009)

Not listed on the Ozone Depletion list (Regulation EU 1005/2009)

Explosives Precursors Regulation (2019/1148)

Contains no substance(s) listed on the Explosives Precursors list (Regulation EU 2019/1148 on the marketing and use of explosives precursors)

Drug Precursors Regulation (273/2004)

Contains no substance(s) listed on the Drug Precursors list (Regulation EC 273/2004 on the manufacture and the placing on market of certain substances used in the illicit manufacture of narcotic drugs and psychotropic substances)

15.1.2. National regulations

California Proposition 65 - This product does not contain any substances known to the state of California to cause cancer, developmental and/or reproductive harm

Germany

Employment restrictions : Observe restrictions according Act on the Protection of Working Mothers (MuSchG). Observe restrictions according Act on the Protection of Young People in Employment (JArbSchG).

Water hazard class (WGK) : Not classified according to Regulation Governing Systems for Handling Substances Hazardous to Waters (AwSV).

Storage class (LGK, TRGS 510) : LGK 13 - Non-combustible solids.

Joint storage table :

LGK 1	LGK 2A	LGK 2B	LGK 3	LGK 4.1A
LGK 4.1B	LGK 4.2	LGK 4.3	LGK 5.1A	LGK 5.1B
LGK 5.1C	LGK 5.2	LGK 6.1A	LGK 6.1B	LGK 6.1C
LGK 6.1D	LGK 6.2	LGK 7	LGK 8A	LGK 8B
LGK 10	LGK 11	LGK 12	LGK 13	LGK 10-13

Joint storage not permitted for : LGK 1, LGK 6.2, LGK 7.

Joint storage with restrictions permitted for : LGK 4.1A, LGK 5.1C.

Joint storage permitted for : LGK 2A, LGK 2B, LGK 3, LGK 4.1B, LGK 4.2, LGK 4.3, LGK 5.1A, LGK 5.1B, LGK 5.2, LGK 6.1A, LGK 6.1B, LGK 6.1C, LGK 6.1D, LGK 8A, LGK 8B, LGK 10, LGK 11, LGK 12, LGK 13, LGK 10-13.

Hazardous Incident Ordinance (12. BImSchV) : Is not subject of the Hazardous Incident Ordinance (12. BImSchV)

Netherlands

ABM category : A(4) - low hazard for aquatic organisms, may have longterm hazardous effects in aquatic environment

SZW-lijst van kankerverwekkende stoffen : The substance is not listed

SZW-lijst van mutagene stoffen : The substance is not listed

SZW-lijst van reprotoxische stoffen – Borstvoeding : The substance is not listed

SZW-lijst van reprotoxische stoffen – Vruchtbaarheid : The substance is not listed

SZW-lijst van reprotoxische stoffen – Ontwikkeling : The substance is not listed

Switzerland

Storage class (LK) : LK 11/13 - Solids

15.2. Chemical safety assessment

No additional information available

SECTION 16: Other information

Full text of H- and EUH-statements:	
Eye Irrit. 2	Serious eye damage/eye irritation, Category 2
H319	Causes serious eye irritation.

Full text of use descriptors	
ERC2	Formulation into mixture
ERC8a	Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor)
ERC9a	Widespread use of functional fluid (indoor)

Full text of use descriptors	
ERC9b	Widespread use of functional fluid (outdoor)
PC18	Ink and Toners
PC39	Cosmetics, personal care products
PROC1	Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions
PROC14	Tabletting, compression, extrusion, pelettisation, granulation
PROC15	Use as laboratory reagent
PROC2	Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions
PROC3	Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition
PROC4	Chemical production where opportunity for exposure arises
PROC5	Mixing or blending in batch processes
PROC8a	Transfer of substance or mixture (charging and discharging) at non-dedicated facilities
PROC8b	Transfer of substance or mixture (charging and discharging) at dedicated facilities
PROC9	Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Safety Data Sheet (SDS), EU - Biorius

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

Annex to the extended Safety Data Sheet (eSDS)

Industrial

Identification of the substance or mixture
Product definition : UVCB
Product name : Sisterna SP70-C

Section 1 - Title

Short title of the exposure scenario : Formulation or re-packing; Cosmetics, personal care products (PC39)

List of use descriptors : **Identified use name: ES01:** Industrial use - Manufacture of cosmetics.: PC39; PROC01, PROC02, PROC03, PROC05, PROC08a, PROC08b, PROC09, PROC14, PROC15; ERC02
Process Category: PROC01, PROC02, PROC03, PROC05, PROC08a, PROC08b, PROC09, PROC14, PROC15
Subsequent service life relevant for that use: No.
Environmental Release Category: ERC02
Market sector by type of chemical product: PC39

Environmental contributing scenarios : **Formulation of low viscosity liquids (small scale) -** ERC02
Formulation of Fine Fragrances - Cleaning with Water - ERC02
Formulation of High Viscosity Body Care Products (small scale) - ERC02
Formulation of Non-liquid Creams (small scale) - ERC02
Formulation of cosmetic products involving cleaning with organic solvents (varnish, removers, decorative cosmetics, spray, lacquer, fine fragrance, solar oil, solid products) (small scale) - ERC02
Formulation of solid cosmetic and home care products (small scale) - ERC02

Health Contributing scenarios : **Contributing scenarios -** PROC01, PROC02, PROC03, PROC05, PROC08a, PROC08b, PROC09, PROC14, PROC15

Number of the ES : 01

Section 2 - Exposure controls

Contributing scenario controlling environmental exposure for 1: Formulation of low viscosity liquids (small scale)

Amounts used : Daily amount per site: ≤0.4 tonnes/day.
Annual amount per site: ≤100 tonnes/year.

Frequency and duration of use : Emission days: 250 days per year.

Other conditions affecting environmental exposure : Receiving surface water flow: ≥18000 m3/d.

Release factor after on-site risk management:
water: 0.008% (Cosmetics Europe SPERC 2.1.c.v2).
air: 0% (Cosmetics Europe SPERC 2.1.c.v2).
Soil: 0% (Cosmetics Europe SPERC 2.1.c.v2).

Conditions and measures related to sewage treatment plant : Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)
Discharge rate: ≥2000 m^3/d.
Application of the STP sludge on agricultural soil: Yes.
On-site wastewater treatment: Yes. (Efficiency of at least 98%)

Conditions and measures related to external treatment of waste for disposal : No waste from process.

Contributing scenario controlling environmental exposure for 2: Formulation of Fine Fragrances - Cleaning with Water

Amounts used : Daily amount per site: ≤0.4 tonnes/day.
Annual amount per site: ≤100 tonnes/year.

Frequency and duration of use : Emission days: 250 days per year.

Other conditions affecting environmental exposure : Release factor after on-site risk management:

water: 0.015% (Cosmetics Europe SPERC 2.1.d.v2).
air: 0% (Cosmetics Europe SPERC 2.1.d.v2).
Soil: 0% (Cosmetics Europe SPERC 2.1.d.v2).

Conditions and measures related to sewage treatment plant : Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)

Discharge rate: ≥2000 m^3/d.
Application of the STP sludge on agricultural soil: No.
On-site wastewater treatment: Yes. (Efficiency of at least 99%)

Conditions and measures related to external treatment of waste for disposal	:	Particular considerations on the waste treatment operations.
Contributing scenario controlling environmental exposure for 3: Formulation of High Viscosity Body Care Products (small scale)		
Amounts used	:	Daily amount per site: ≤0.2 tonnes/day. Annual amount per site: ≤50 tonnes/year.
Frequency and duration of use	:	Emission days: 250 days per year.
Other conditions affecting environmental exposure	:	Receiving surface water flow: ≥18000 m³/d. Release factor after on-site risk management: water: 0.02% (Cosmetics Europe SPERC 2.1.g.v2). air: 0% (Cosmetics Europe SPERC 2.1.g.v2). Soil: 0% (Cosmetics Europe SPERC 2.1.g.v2).
Technical conditions and measures at process level (source) to prevent release	:	Oil-water separation.
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%) Discharge rate: ≥2000 m³/d. Application of the STP sludge on agricultural soil: No. On-site wastewater treatment: Yes. (Efficiency of at least 99%)
Conditions and measures related to external treatment of waste for disposal	:	Particular considerations on the waste treatment operations.
Contributing scenario controlling environmental exposure for 4: Formulation of Non-liquid Creams (small scale)		
Amounts used	:	Daily amount per site: ≤0.1 tonnes/day. Annual amount per site: ≤25 tonnes/year.
Frequency and duration of use	:	Emission days: 250 days per year.
Other conditions affecting environmental exposure	:	Release factor after on-site risk management: water: 0.04% (Cosmetics Europe SPERC 2.1.j.v2). air: 0% (Cosmetics Europe SPERC 2.1.j.v2). Soil: 0% (Cosmetics Europe SPERC 2.1.j.v2).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%) Discharge rate: ≥2000 m³/d. Application of the STP sludge on agricultural soil: No. On-site wastewater treatment: Yes. (Efficiency of at least 99%)
Conditions and measures related to external treatment of waste for disposal	:	Particular considerations on the waste treatment operations.

Contributing scenario controlling environmental exposure for 5: Formulation of cosmetic products involving cleaning with organic solvents (varnish, removers, decorative cosmetics, spray, lacquer, fine fragrance, solar oil, solid products) (small scale) **Contributing scenario controlling environmental exposure for 6: Formulation of solid cosmetic and home care products (small scale)**		
Amounts used	:	Daily amount per site: ≤0.4 tonnes/day. Annual amount per site: ≤100 tonnes/year.
Frequency and duration of use	:	Emission days: 250 days per year.
Other conditions affecting environmental exposure	:	Receiving surface water flow: ≥18000 m³/d. Release factor after on-site risk management: water: 0% (Cosmetics Europe SPERC 2.2.c.v2). air: 0% (Cosmetics Europe SPERC 2.2.c.v2). Soil: 0% (Cosmetics Europe SPERC 2.2.c.v2).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%) Discharge rate: ≥2000 m³/d. Application of the STP sludge on agricultural soil: Yes.
Conditions and measures related to external treatment of waste for disposal	:	Particular considerations on the waste treatment operations
Contributing scenario controlling environmental exposure for 6: Formulation of solid cosmetic and home care products (small scale)		

Amounts used	:	Daily amount per site: ≤0.4 tonnes/day. Annual amount per site: ≤100 tonnes/year.
Frequency and duration of use	:	Emission days: 250 days per year.
Other conditions affecting environmental exposure	:	Receiving surface water flow: ≥18000 m^3/d. Release factor after on-site risk management: water: 0.01% (Cosmetics Europe / AISE SPERC 2.3.c.v2). air: 0% (Cosmetics Europe / AISE SPERC 2.3.c.v2). Soil: 0% (Cosmetics Europe / AISE SPERC 2.3.c.v2).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%) Discharge rate: ≥2000 m^3/d. Application of the STP sludge on agricultural soil: Yes. On-site wastewater treatment: Yes. (Efficiency of at least 95%)
Conditions and measures related to external treatment of waste for disposal	:	Particular considerations on the waste treatment operations

Contributing scenario controlling worker exposure for: All Contributing scenarios

Concentration of substance in mixture or article	:	Covers percentage substance in the product up to 9%. PROC08a, PROC08b: Covers percentage substance in the product up to 100%.
Physical state	:	Solid.
Frequency and duration of use/exposure	:	Covers daily exposures up to 8 hours.
Other conditions affecting workers exposure	:	Indoor use. Process temperature: 40 °C.
Ventilation control measures	:	Provide a basic standard of general ventilation (1 to 3 air changes per hour).
Organisational measures to prevent/limit releases, dispersion and exposure	:	PROC08a, PROC08b: - Minimisation of manual phases. - Ensure that splashes and spills are minimised. - Avoid contact with contaminated tools and objects. - Regular cleaning of equipment and work area. - Management/supervision in place to check that the RMMs in place are being used correctly and OCs followed. - Training for staff on good practice. - Good standard of personal hygiene.

Conditions and measures related to personal protection, hygiene and health evaluation

Personal protection	:	PROC08a, PROC08b: Wear chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact.

Contributing scenario controlling worker exposure for 7: Contributing scenarios

CS07: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions - PROC01
CS08: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions - PROC02
CS09: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition - PROC03
CS10: Mixing or blending in batch processes - PROC05
CS11: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities - PROC08a
CS12:Transfer of substance or mixture (charging and discharging) at dedicated facilities - PROC08b
CS13: Transfer of substance or mixture into small containers (dedicated filling line, including weighing) - PROC09
CS14: Tabletting, compression, extrusion, pelletisation, granulation - PROC14
CS15: Use as laboratory reagent - PROC15

Section 3 - Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment: 1: Formulation of low viscosity liquids (small scale)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.000182 mg/l. Risk characterisation ratio (PEC/PNEC): 0.055.

		Freshwater sediment: 0.22 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.552. Marine water: 0.000018 mg/l. Risk characterisation ratio (PEC/PNEC): 0.054. Marine water sediment: 0.022 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.545. Sewage Treatment Plant: 0.00128 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.051 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.642.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).
Exposure estimation and reference to its source - Environment: 2: Formulation of Fine Fragrances - Cleaning with Water		
Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.000292 mg/l. Risk characterisation ratio (PEC/PNEC): 0.088. Freshwater sediment: 0.352 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.885. Marine water: 0.000029 mg/l. Risk characterisation ratio (PEC/PNEC): 0.088. Marine water sediment: 0.035 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.878. Sewage Treatment Plant: 0.0024 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01.1. Soil: 0.00126 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.016.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).
Exposure estimation and reference to its source - Environment: 3: Formulation of High Viscosity Body Care Products (small scale)		
Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.000213 mg/l. Risk characterisation ratio (PEC/PNEC): 0.065. Freshwater sediment: 0.257 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.647. Marine water: 0.0000211 mg/l. Risk characterisation ratio (PEC/PNEC): 0.064. Marine water sediment: 0.025 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.64. Sewage Treatment Plant: 0.0016 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.00126 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.016.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).
Exposure estimation and reference to its source - Environment: 4: Formulation of Non-liquid Creams (small scale)		
Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.000213 mg/l. Risk characterisation ratio (PEC/PNEC): 0.065. Freshwater sediment: 0.257 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.647. Marine water: 0.0000211 mg/l. Risk characterisation ratio (PEC/PNEC): 0.064. Marine water sediment: 0.025 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.64. Sewage Treatment Plant: 0.0016 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.00126 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.016.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Environment: 5: Formulation of cosmetic products involving cleaning with organic solvents (varnish, removers, decorative cosmetics, spray, lacquer, fine fragrance, solar oil, solid products) (small scale)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000565 mg/l. Risk characterisation ratio (PEC/PNEC): 0.017. Freshwater sediment: 0.068 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.171. Marine water: 0.00000542 mg/l. Risk characterisation ratio (PEC/PNEC): 0.016. Marine water sediment: 0.00653 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.164. Sewage Treatment Plant: 0 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.00126 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.016.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Environment: 6: Formulation of solid cosmetic and home care products (small scale)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.000213 mg/l. Risk characterisation ratio (PEC/PNEC): 0.065. Freshwater sediment: 0.257 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.647. Marine water: 0.0000211 mg/l. Risk characterisation ratio (PEC/PNEC): 0.064. Marine water sediment: 0.025 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.64. Sewage Treatment Plant: 0.0016 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.063 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.798.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Workers: 7: Contributing scenarios

Exposure assessment (human)	:	According to the Classification, Labelling and Packaging of substances Regulation (EC) No 272/2008, mixtures containing < 10% of the registered substance do not meet the criteria for classification as Eye Irritant Cat 2. It is therefore considered that there is no unacceptable risk to workers from any eye exposure to Fatty acids C16-18 (even numbered), mono diesters with sucrose during formulation. PROC08a, PROC08b: Qualitative approach used to conclude safe use.
Exposure estimation	:	Not applicable.

Section 4 - Guidance to DU to evaluate whether he works inside the boundaries set by the ES

General	:	The immediate downstream user is required to evaluate whether the operational conditions and risk management measures described in the exposure scenario fit to his use. If other OC/RMM are adopted, the user has to ensure that risks are managed to at least equivalent levels. The risk assessment methods/tools given in section 3 may be used for this evaluation.
Environment	:	Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required.

Annex to the extended Safety Data Sheet (eSDS)

Professional

Identification of the substance or mixture		
Product definition	:	UVCB
Product name	:	Sisterna SP70-C
Section 1 - Title		
Short title of the exposure scenario	:	Widespread use by professional workers; Cosmetics, personal care products (PC39)
List of use descriptors	:	**Identified use name: ES0**2: Widespread use by professional workers - Hairdressing services: PC39; PROC05, PROC08a; ERC08a **Process Category:** PROC05, PROC08a **Subsequent service life relevant for that use:** No. **Environmental Release Category:** ERC08a **Market sector by type of chemical product:** PC39
Environmental contributing scenarios	:	**Wide Dispersive Use in 'Down the Drain' products - hair and skin care products (Consumers and Professionals) -** ERC08a **Wide Dispersive Use in Aerosol products for hair and skin care (Propellants) -** ERC08a **Wide Dispersive Use of Aerosol products for hair and skin care (Non-Propellants) -** ERC08a
Health Contributing scenarios	:	**Contributing scenarios -** PROC05, PROC08a
Number of the ES	:	02

Section 2 - Exposure controls

Contributing scenario controlling environmental exposure for 1: Wide Dispersive Use in 'Down the Drain' products - hair and skin care products (Consumers and Professionals)

Amounts used	:	Daily local widespread use amount: ≤0.000011 tonnes/day. Fraction of EU tonnage used in region: 0.053. Fraction of main source to local environment: 0.00075.
Frequency and duration of use	:	Emission days: 365 days per year.
Other conditions affecting environmental exposure	:	Release factor after on-site risk management: water: 100% (Cosmetics Europe SPERC 8a.1.a.v2). air: 0% (Cosmetics Europe SPERC 8a.1.a.v2). Soil: 0% (Cosmetics Europe SPERC 8a.1.a.v2).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)
Conditions and measures related to external treatment of waste for disposal	:	Particular risks from waste treatment unlikely due to small fraction of used substance entering into the waste stage. Waste disposal according to national/local legislation is sufficient.

Contributing scenario controlling environmental exposure for 2: Wide Dispersive Use in Aerosol products for hair and skin care (Propellants)

Amounts used	:	Daily local widespread use amount: ≤0.000011 tonnes/day. Fraction of EU tonnage used in region: 0.053. Fraction of main source to local environment: 0.00075.
Frequency and duration of use	:	Emission days: 365 days per year.
Other conditions affecting environmental exposure	:	Release factor after on-site risk management: water: 0% (Cosmetics Europe SPERC 8a.1.b.v2). air: 100% (Cosmetics Europe SPERC 8a.1.b.v2). Soil: 0% (Cosmetics Europe SPERC 8a.1.b.v2).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)
Conditions and measures related to external treatment of waste for disposal	:	Particular risks from waste treatment unlikely due to small fraction of used substance entering into the waste stage. Waste disposal according to national/local legislation is sufficient.

Contributing scenario controlling environmental exposure for 3: Wide Dispersive Use of Aerosol products for hair and skin care (Non-Propellants)

Amounts used	:	Daily local widespread use amount: ≤0.000011 tonnes/day. Fraction of EU tonnage used in region: 0.053. Fraction of main source to local environment: 0.00075.
Frequency and duration of use	:	Emission days: 365 days per year.
Other conditions affecting environmental exposure	:	Release factor after on-site risk management: water: 100% (Cosmetics Europe SPERC 8a.1.c.v2). air: 0% (Cosmetics Europe SPERC 8a.1.c.v2). Soil: 0% (Cosmetics Europe SPERC 8a.1.c.v2).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)

Conditions and measures related to external treatment of waste for disposal	:	Particular risks from waste treatment unlikely due to small fraction of used substance entering into the waste stage. Waste disposal according to national/local legislation is sufficient.

Contributing scenario controlling worker exposure for: All Contributing scenarios

Concentration of substance in mixture or article	:	Covers percentage substance in the product up to 9%.
Physical state	:	Liquid.
Frequency and duration of use/exposure	:	Covers daily exposures up to 8 hours.
Other conditions affecting workers exposure	:	Indoor use. Process temperature: 40 °C.
Ventilation control measures	:	Provide a basic standard of general ventilation (1 to 3 air changes per hour).

Contributing scenario controlling worker exposure for 4: Contributing scenarios

CS04: Mixing or blending in batch processes - PROC05
CS05: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities - PROC08a

Section 3 - Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment: 1: Wide Dispersive Use in 'Down the Drain' products - hair and skin care products (Consumers and Professionals)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000997 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Freshwater sediment: 0.12 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.302. Marine water: 0.00000973 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Marine water sediment: 0.012 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.295. Sewage Treatment Plant: 0.000439 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.018 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.231.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Environment: 2: Wide Dispersive Use in Aerosol products for hair and skin care (Propellants)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000565 mg/l. Risk characterisation ratio (PEC/PNEC): 0.017. Freshwater sediment: 0.068 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.171. Marine water: 0.00000542 mg/l. Risk characterisation ratio (PEC/PNEC): 0.016. Marine water sediment: 0.00653 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.164. Sewage Treatment Plant: 0 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.00126 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.016.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Environment: 3: Wide Dispersive Use of Aerosol products for hair and skin care (Non-Propellants)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000997 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Freshwater sediment: 0.12 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.302. Marine water: 0.00000973 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Marine water sediment: 0.012 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.295.

		Sewage Treatment Plant: 0.000439 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.018 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.231.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Workers: 4: Contributing scenarios		
Exposure assessment (human)	:	According to the Classification, Labelling and Packaging of substances Regulation (EC) No 272/2008, mixtures containing < 10% of the registered substance do not meet the criteria for classification as Eye Irritant Cat 2. It is therefore considered that there is no unacceptable risk to workers from any eye exposure to Fatty acids C16-18 (even numbered), mono diesters with sucrose during use of cosmetics.
Exposure estimation	:	Not applicable.

Section 4 - Guidance to DU to evaluate whether he works inside the boundaries set by the ES

General	:	The immediate downstream user is required to evaluate whether the operational conditions and risk management measures described in the exposure scenario fit to his use. If other OC/RMM are adopted, the user has to ensure that risks are managed to at least equivalent levels. The risk assessment methods/tools given in section 3 may be used for this evaluation.
Environment	:	Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required.

Annex to the extended Safety Data Sheet (eSDS)

Industrial

Identification of the substance or mixture

Product definition : UVCB

Product name : Sisterna SP70-C

Section 1 - Title

Short title of the exposure scenario : Formulation or re-packing; Ink and toners (PC18)

List of use descriptors : **Identified use name: ES03:** Industrial use - Formulation of non-cosmetic products (inks): PC18; PROC01, PROC02, PROC03, PROC04, PROC05, PROC08a, PROC09; ERC02
Process Category: PROC01, PROC02, PROC03, PROC04, PROC05, PROC08a, PROC09
Subsequent service life relevant for that use: No.
Environmental Release Category: ERC02
Market sector by type of chemical product: PC18

Environmental contributing scenarios : **Formulation of non-cosmetic products (inks) -** ERC02

Health Contributing scenarios : **Contributing scenarios -** PROC01, PROC02, PROC03, PROC04, PROC05, PROC08a, PROC09

Number of the ES : 03

Section 2 - Exposure controls

Contributing scenario controlling environmental exposure for 1: Formulation of non-cosmetic products (inks)

Amounts used : Daily amount per site: ≤0.44 tonnes/day.
Annual amount per site: ≤100 tonnes/year.

Frequency and duration of use : Emission days: 225 days per year.

Other conditions affecting environmental exposure : Receiving surface water flow: ≥18000 m^3/d.

Conditions and measures related to sewage treatment plant : Release factor after on-site risk management:
water: 0.005% (CEPE SPERC 2.3a.v1).
air: 0.0097% (CEPE SPERC 2.3a.v1).
Soil: 0% (CEPE SPERC 2.3a.v1).
Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)
Discharge rate: ≥2000 m^3/d.
Application of the STP sludge on agricultural soil: Yes.

Conditions and measures related to external treatment of waste for disposal : Particular considerations on the waste treatment operations.

Contributing scenario controlling worker exposure for: All Contributing scenarios

Concentration of substance in mixture or article : Covers percentage substance in the product up to 9%.
PROC04, PROC08a: Covers percentage substance in the product up to 100%.

Physical state : Liquid/ Solid.

Frequency and duration of use/exposure : Covers daily exposures up to 8 hours.

Other conditions affecting workers exposure : Indoor use.
Process temperature: 40 °C.

Ventilation control measures : Provide a basic standard of general ventilation (1 to 3 air changes per hour).

Organisational measures to prevent/limit releases, dispersion and exposure : PROC04, PROC08a:
- Minimisation of manual phases.
- Ensure that splashes and spills are minimised.
- Avoid contact with contaminated tools and objects.
- Regular cleaning of equipment and work area.
- Management/supervision in place to check that the RMMs in place are being used correctly and OCs followed.
- Training for staff on good practice.
- Good standard of personal hygiene.

Conditions and measures related to personal protection, hygiene and health evaluation

Personal protection : PROC04, PROC08a:
Wear chemically resistant face shield, goggles or safety glasses with side shields when there is potential for direct contact.

Contributing scenario controlling worker exposure for 2: Contributing scenarios

CS02: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions - PROC01
CS03: Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions - PROC02
CS04: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition - PROC03
CS05: Chemical production where opportunity for exposure arises - PROC04
CS06: Mixing or blending in batch processes - PROC05
CS07: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities - PROC08a

CS08: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities - PROC08a
CS09: Transfer of substance or mixture into small containers (dedicated filling line, including weighing) - PROC09

Section 3 - Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment: 1: Formulation of non-cosmetic products (inks)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.000143 mg/l. Risk characterisation ratio (PEC/PNEC): 0.043. Freshwater sediment: 0.172 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.433. Marine water: 0.000014 mg/l. Risk characterisation ratio (PEC/PNEC): 0.043. Marine water sediment: 0.017 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.426. Sewage Treatment Plant: 0.000879 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.036 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.448.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Workers: 2: Contributing scenarios

Exposure assessment (human)	:	According to the Classification, Labelling and Packaging of substances Regulation (EC) No 272/2008, mixtures containing < 10% of the registered substance do not meet the criteria for classification as Eye Irritant Cat 2. It is therefore considered that there is no unacceptable risk to workers from any eye exposure to Fatty acids C16-18 (even numbered), mono diesters with sucrose during formulation. PROC04, PROC08a: Qualitative approach used to conclude safe use.
Exposure estimation	:	Not applicable.

Section 4 - Guidance to DU to evaluate whether he works inside the boundaries set by the ES

General	:	The immediate downstream user is required to evaluate whether the operational conditions and risk management measures described in the exposure scenario fit to his use. If other OC/RMM are adopted, the user has to ensure that risks are managed to at least equivalent levels. The risk assessment methods/tools given in section 3 may be used for this evaluation.
Environment	:	Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required.

Annex to the extended Safety Data Sheet (eSDS)

Consumer

Identification of the substance or mixture

Product definition : UVCB

Product name : Sisterna SP70-C

Section 1 - Title

Short title of the exposure scenario : Consumer use; Cosmetics, personal care products (PC39)

List of use descriptors : **Identified use name: ES04:** Consumer use - End use of cosmetic products: PC39; ERC08a
Subsequent service life relevant for that use: No.
Environmental Release Category: ERC08a
Market sector by type of chemical product: PC39

Environmental contributing scenarios : **Wide Dispersive Use in 'Down the Drain' products - hair and skin care products (Consumers and Professionals) -** ERC08a
Wide Dispersive Use in Aerosol products for hair and skin care (Propellants) - ERC08a
Wide Dispersive Use of Aerosol products for hair and skin care (Non-Propellants) - ERC08a

Health Contributing scenarios : **Cosmetics, personal care products -** PC39

Number of the ES : 04

Section 2 - Exposure controls

Contributing scenario controlling environmental exposure for 1: Wide Dispersive Use in 'Down the Drain' products - hair and skin care products (Consumers and Professionals)

Amounts used : Daily local widespread use amount: ≤0.000011 tonnes/day.
Fraction of EU tonnage used in region: 0.053.
Fraction of main source to local environment: 0.00075.

Frequency and duration of use : Emission days: 365 days per year.

Other conditions affecting environmental exposure : Release factor after on-site risk management:
water: 100% (Cosmetics Europe SPERC 8a.1.a.v2).
air: 0% (Cosmetics Europe SPERC 8a.1.a.v2).
Soil: 0% (Cosmetics Europe SPERC 8a.1.a.v2).

Conditions and measures related to sewage treatment plant : Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)

Conditions and measures related to external treatment of waste for disposal : Particular risks from waste treatment unlikely due to small fraction of used substance entering into the waste stage. Waste disposal according to national/local legislation is sufficient.

Contributing scenario controlling environmental exposure for 2: Wide Dispersive Use in Aerosol products for hair and skin care (Propellants)

Amounts used : Daily local widespread use amount: ≤0.000011 tonnes/day.
Fraction of EU tonnage used in region: 0.053.
Fraction of main source to local environment: 0.00075.

Frequency and duration of use : Emission days: 365 days per year.

Other conditions affecting environmental exposure : Release factor after on-site risk management:
water: 0% (Cosmetics Europe SPERC 8a.1.b.v2).
air: 100% (Cosmetics Europe SPERC 8a.1.b.v2).
Soil: 0% (Cosmetics Europe SPERC 8a.1.b.v2).

Conditions and measures related to sewage treatment plant : Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)

Conditions and measures related to external treatment of waste for disposal : Particular risks from waste treatment unlikely due to small fraction of used substance entering into the waste stage. Waste disposal according to national/local legislation is sufficient.

Contributing scenario controlling environmental exposure for 3: Wide Dispersive Use of Aerosol products for hair and skin care (Non-Propellants)

Amounts used : Daily local widespread use amount: ≤0.000011 tonnes/day.
Fraction of EU tonnage used in region: 0.053.
Fraction of main source to local environment: 0.00075.

Frequency and duration of use : Emission days: 365 days per year.

Other conditions affecting environmental exposure : Release factor after on-site risk management:
water: 100% (Cosmetics Europe SPERC 8a.1.c.v2).
air: 0% (Cosmetics Europe SPERC 8a.1.c.v2).
Soil: 0% (Cosmetics Europe SPERC 8a.1.c.v2).

Conditions and measures related to sewage treatment plant : Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)

Conditions and measures related to external treatment of waste for disposal	:	Particular risks from waste treatment unlikely due to small fraction of used substance entering into the waste stage. Waste disposal according to national/local legislation is sufficient.

Contributing scenario controlling consumer exposure for 4: Cosmetics, personal care products

Concentration of substance in mixture or article	:	Covers percentage substance in the product up to 9%.
Physical state	:	Solid.
Frequency and duration of use/exposure	:	Covers daily exposures up to 8 hours.

Section 3 - Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment: 1: Wide Dispersive Use in 'Down the Drain' products - hair and skin care products (Consumers and Professionals)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000997 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Freshwater sediment: 0.12 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.302. Marine water: 0.00000973 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Marine water sediment: 0.012 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.295. Sewage Treatment Plant: 0.000439 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.018 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.231.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Environment: 2: Wide Dispersive Use in Aerosol products for hair and skin care (Propellants)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000565 mg/l. Risk characterisation ratio (PEC/PNEC): 0.017. Freshwater sediment: 0.068 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.171. Marine water: 0.00000542 mg/l. Risk characterisation ratio (PEC/PNEC): 0.016. Marine water sediment: 0.00653 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.164. Sewage Treatment Plant: 0 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.00126 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.016.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Environment: 3: Wide Dispersive Use of Aerosol products for hair and skin care (Non-Propellants)

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000997 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Freshwater sediment: 0.12 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.302. Marine water: 0.00000973 mg/l. Risk characterisation ratio (PEC/PNEC): 0.03. Marine water sediment: 0.012 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.295. Sewage Treatment Plant: 0.000439 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.018 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.231.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Consumers: 4: Cosmetics, personal care products

Exposure assessment (human)	:	According to the Classification, Labelling and Packaging of substances Regulation (EC) No 272/2008, mixtures containing < 10% of the registered substance do not meet the criteria for classification as Eye Irritant Cat 2. It is therefore considered that there is no unacceptable risk to consumers from any eye exposure to Fatty acids C16-18 (even numbered), mono diesters with sucrose during use.
Exposure estimation	:	Not applicable.

Section 4 - Guidance to DU to evaluate whether he works inside the boundaries set by the ES

General	:	The immediate downstream user is required to evaluate whether the operational conditions and risk management measures described in the exposure scenario fit to his use. If other OC/RMM are adopted, the user has to ensure that risks are managed to at least equivalent levels. The risk assessment methods/tools given in section 3 may be used for this evaluation.
Environment	:	Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required.

Annex to the extended Safety Data Sheet (eSDS)

Consumer

Identification of the substance or mixture

Product definition	:	UVCB
Product name	:	Sisterna SP70-C

Section 1 - Title

Short title of the exposure scenario	:	Consumer use; Ink and toners (PC39)
List of use descriptors	:	**Identified use name: ES05:** Consumer use - Ink products: PC18; ERC09a, ERC09b **Subsequent service life relevant for that use:** No. **Environmental Release Category:** ERC09a, ERC09b **Market sector by type of chemical product:** PC18
Environmental contributing scenarios	:	**Ink products -** ERC09a, ERC09b
Health Contributing scenarios	:	**Ink products -** PC18

Number of the ES	:	05

Section 2 - Exposure controls

Contributing scenario controlling environmental exposure for 1: Ink products

Amounts used	:	Daily local widespread use amount: ≤0.000055 tonnes/day.
Other conditions affecting environmental exposure	:	Release factor after on-site risk management: water: 5% (ERC09b). air: 5% (ERC09b). Soil: 5% (ERC09b).
Conditions and measures related to sewage treatment plant	:	Sewage Treatment Plant: Yes. (Efficiency of at least 92.01%)
Conditions and measures related to external treatment of waste for disposal	:	Particular considerations on the waste treatment operations.

Contributing scenario controlling consumer exposure for 2: Ink products

Concentration of substance in mixture or article	:	Covers percentage substance in the product up to 9%.
Physical state	:	Liquid.

Section 3 - Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment: 1: Ink products

Exposure assessment (environment)	:	EUSES 2.1.2
Exposure estimation	:	Freshwater: 0.0000673 mg/l. Risk characterisation ratio (PEC/PNEC): 0.02. Freshwater sediment: 0.081 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.204. Marine water: 0.0000065 mg/l. Risk characterisation ratio (PEC/PNEC): 0.02. Marine water sediment: 0.00783 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.197. Sewage Treatment Plant: 0.00011 mg/l. Risk characterisation ratio (PEC/PNEC): <0.01. Soil: 0.00553 mg/kg dwt. Risk characterisation ratio (PEC/PNEC): 0.07.
Remark	:	Based on the applied RMMs the risk towards environment is sufficiently controlled (RCR < 1).

Exposure estimation and reference to its source - Consumers: 2: Ink products

Exposure assessment (human)	:	No exposure assessment presented for human health.
Exposure estimation	:	Not applicable.

Section 4 - Guidance to DU to evaluate whether he works inside the boundaries set by the ES

General	:	The immediate downstream user is required to evaluate whether the operational conditions and risk management measures described in the exposure scenario fit to his use. If other OC/RMM are adopted, the user has to ensure that risks are managed to at least equivalent levels. The risk assessment methods/tools given in section 3 may be used for this evaluation.
Environment	:	Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required.